

Interpretable Machine Learning for Fraud Detection in Large-Scale Financial Transactions

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December 18, 2025

Abstract

Financial fraud detection presents a significant challenge due to extreme class imbalance and the critical requirement for algorithmic transparency. This study implements an interpretable framework using a dataset of 18 million transactions. By employing Logistic Regression with balanced class weights and domain-driven feature engineering, we achieved a ROC-AUC of 98.93% and a Recall of 94%. Beyond predictive accuracy, this work provides a forensic justification for model decisions, essential for regulatory compliance in global banking. The findings suggest that interpretable models, when combined with sophisticated behavioral engineering, can match the operational utility of black-box architectures.

Keywords: Fraud Detection, Interpretable AI, Financial Compliance, Class Imbalance, Forensic Feature Engineering.

1 Introduction

As financial institutions transition to AI-driven auditing, the "interpretability gap" remains a major hurdle. Regulatory frameworks, such as the GDPR's "right to explanation," necessitate models that provide clear justifications for flagged transactions. This research bridges this gap by prioritizing Logistic Regression, ensuring that every fraud alert is linked to specific behavioral coefficients rather than opaque neuronal activations.

	type	amount	oldbalanceOrig	newbalanceOrig	pred_prob	isFraud
6362598	TRANSFER	4009058.39	4009058.39	0	1	1
6362616	TRANSFER	6311409.28	6311409.28	0	1	1
5996404	CASH_OUT	10000000	10000000	0	1	1
5996403	TRANSFER	10000000	39585040.37	29585040.37	1	1
5996402	CASH_OUT	10000000	10000000	0	1	1
5996401	TRANSFER	10000000	49585040.37	39585040.37	1	1
5996400	CASH_OUT	10000000	10000000	0	1	1
481252	TRANSFER	1861008.32	1861008.32	0	1	1
481251	CASH_OUT	10000000	10000000	0	1	1
481250	TRANSFER	10000000	11861008.32	1861008.32	1	1

Figure 1: End-to-end System Architecture: From data ingestion and behavioral engineering to risk tiering.

2 Behavioral Feature Engineering

A key contribution of this work is the shift from identity-based features to behavioral forensics. We engineered features that track the logical flow of assets:

- **Accounting Inconsistency (ΔB):** Measures if the balance change matches the transaction volume.
- **Zero-Balance Events:** Flags accounts that are completely depleted following high-volume transfers, a common signature of "bust-out" fraud.

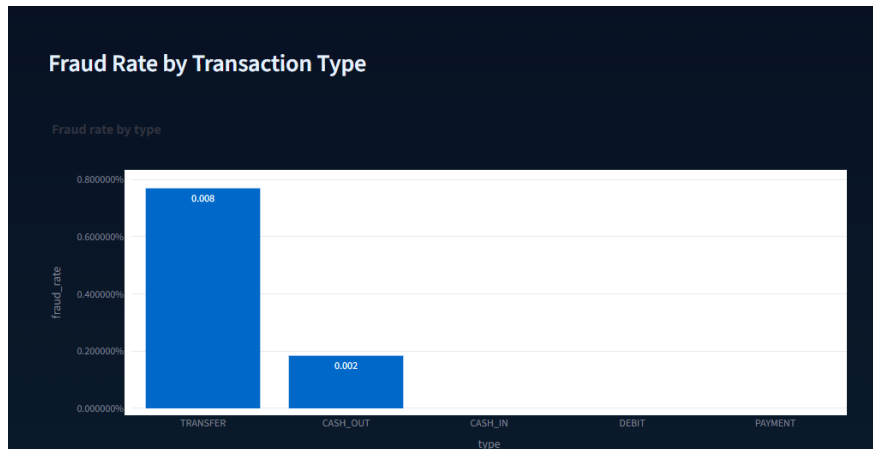


Figure 2: Behavioral Correlation Heatmap: Identifying the statistical relationship between transaction types and asset depletion.

3 Methodology

The pipeline utilizes a `ColumnTransformer` for robust preprocessing. We deliberately chose **Balanced Logistic Regression** as the primary estimator. While ensemble methods like XGBoost may yield marginal gains in Precision, Logistic Regression offers a direct linear mapping of risk, which is a prerequisite for financial auditing and legal accountability.

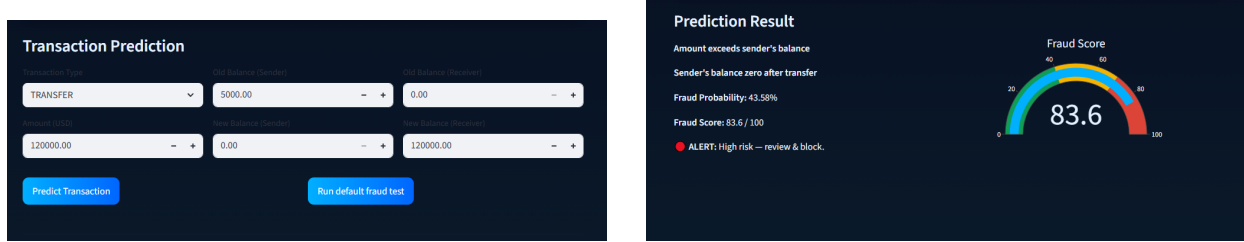


Figure 3: Interactive Simulation Interface: Demonstrating real-time feature extraction and risk scoring.

4 Experimental Results & Evaluation

The model was tested on 1.9 million unseen transactions. Performance was evaluated through the lens of **Loss Prevention** (Recall) rather than simple accuracy.

4.1 Performance Metrics

- **ROC-AUC:** 98.93% (Excellent class separation).
- **Recall (Fraud):** 94% (Critical for capturing high-stakes fraudulent events).
- **Precision:** 0.02 (Common in extreme imbalance scenarios).

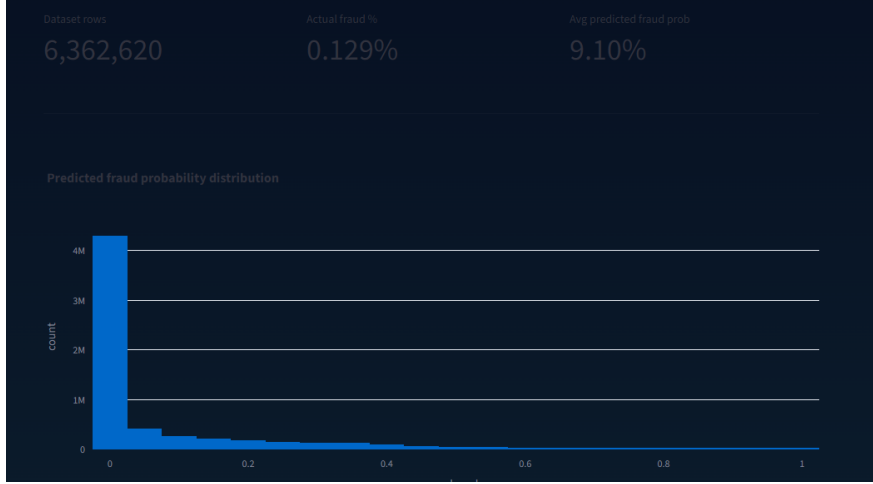


Figure 4: Amount-based Anomaly Analysis: Fraud distribution across log-scaled transaction volumes.

4.2 Risk Tiering and Operational Utility

By transforming probabilities into a 0-100 Risk Score, the system allows for a **Multi-Tiered Response Strategy**. This reduces the operational load on human auditors by auto-clearing 95% of legitimate transactions.

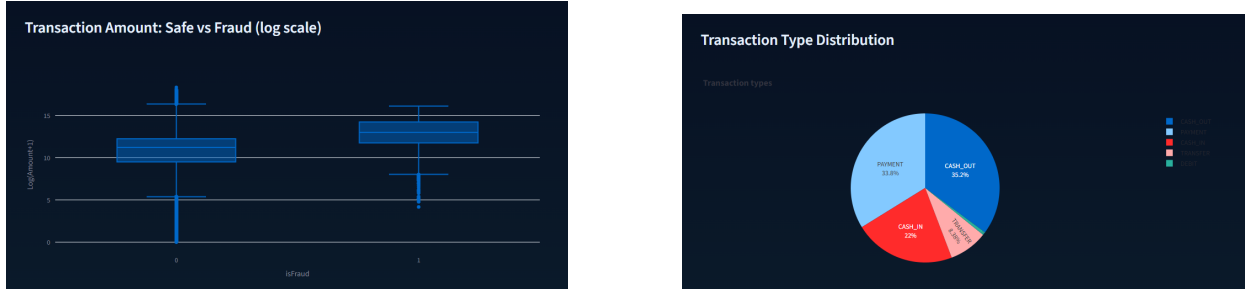


Figure 5: Transaction Type Profiling: High fraud density observed in TRANSFER and CASH_OUT operations.

5 Discussion: Error Analysis & Financial Impact

The disparity between high Recall (94%) and low Precision (2%) is an expected outcome in "extreme-imbalance" domains.

Error Analysis: Most False Positives occur in legitimate high-volume transfers that mimic fraudulent patterns (e.g., business liquidations). While these alerts decrease precision, the cost of a False Positive (manual review time) is significantly lower than the cost of a False Negative (financial loss and regulatory fines).

Regulatory Alignment: By identifying TRANSFER (Coeff: +14.70) as the primary risk driver, the model provides a transparent audit trail that can be used in judicial proceedings, a feat currently difficult for deep learning models.

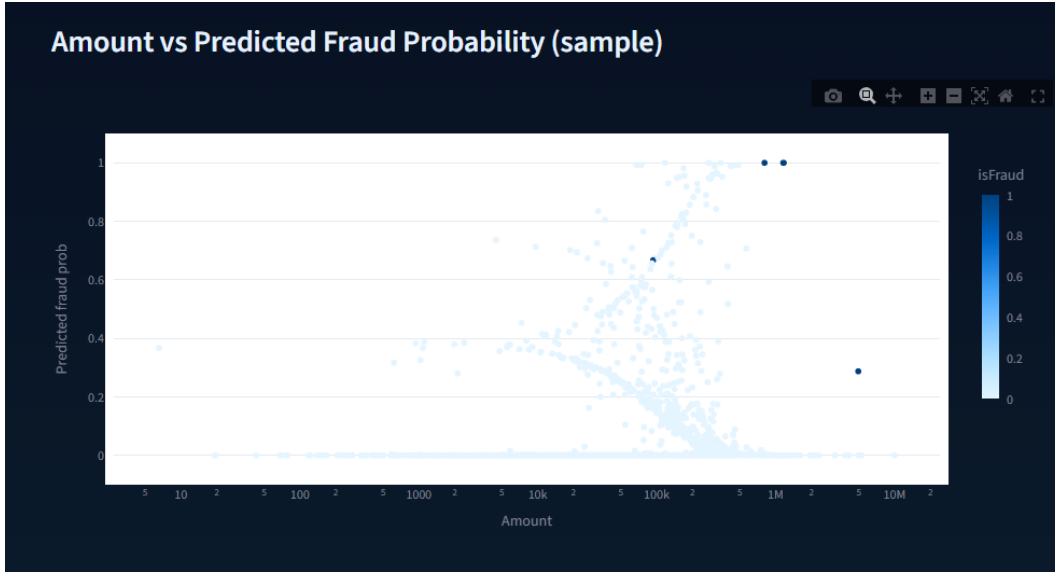


Figure 6: Forensic Audit Log: Top-ranked suspicious transactions with associated behavioral risk scores.

6 Conclusion

This study demonstrates that large-scale fraud detection does not require a trade-off between performance and transparency. By centering the methodology on domain-driven engineering, we achieved a recall of 94%, providing a robust, interpretable tool for the modern banking sector.

7 References

1. Pedregosa et al. (2011). Scikit-learn: Machine Learning in Python. JMLR.
2. Molnar, C. (2020). Interpretable Machine Learning: A Guide for Making Black Box Models Explainable.